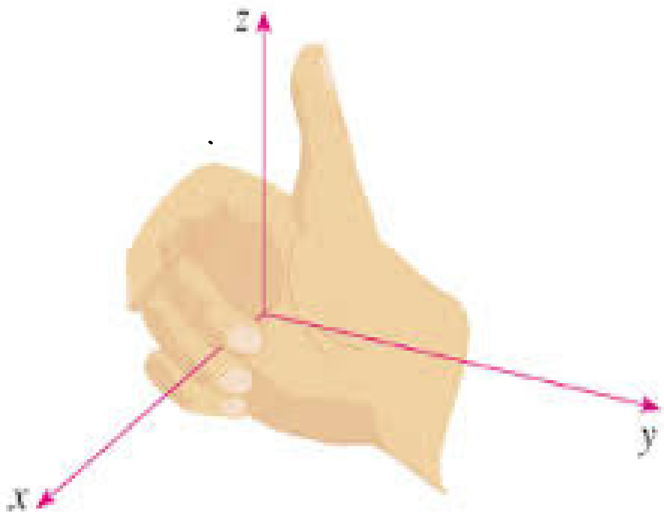
6.1 Three-dimensional coordinate systems

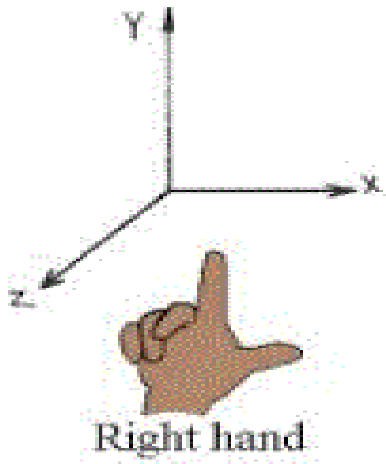
1-dim : $\mathbb{R}^1$ – a line with point a on the line

2-dim : $\mathbb{R}^2$ – a plane with point (a,b) on the plane

3-dim : $\mathbb{R}^3$ – a space with point (a,b,c) in the space

Axes orientation according to the right hand rule.

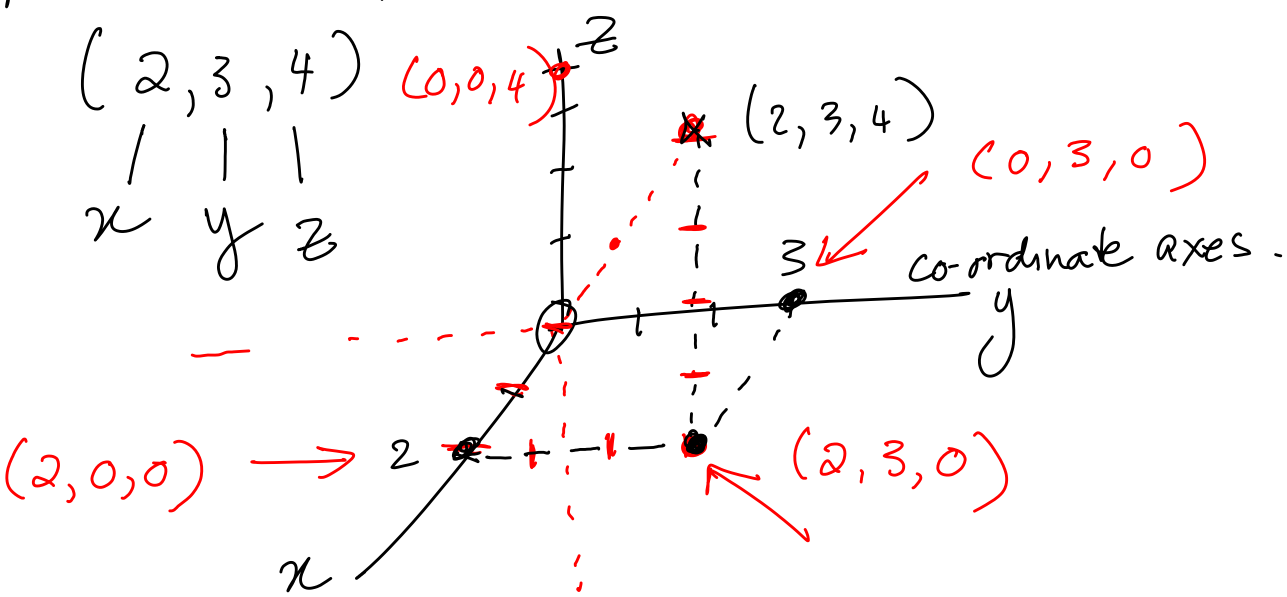




Example of a point in space:

Point $(2, 3, 4)$

$x \quad y \quad z$



Planes in space:

Visualise:

$z = 0$ — xy plane

$y = 0$ — xz plane

$x = 3$ — plane parallel to yz-plane, 3 units forward.

$z = -1$ — plane parallel to xy-plane, dropped one unit

Distance between two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$:

$$|PQ| = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2 + (z_2-z_1)^2}$$

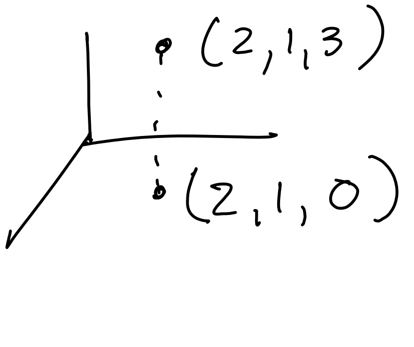
Examples:

1. Distance between $A(1,2,-3)$ and $B(4,0,-3)$

$$|AB| = \sqrt{(4-1)^2 + (0-2)^2 + (-3-(-3))^2} = \sqrt{13}.$$

$|AB|$ → length of line AB

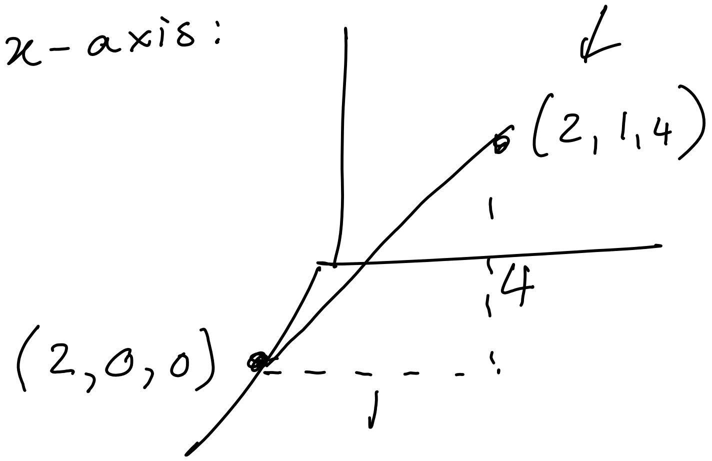
2. Distance between $A(2,1,3)$ and the xy-plane



Distance: 3.

3.

Find the distance from the point $(2,1,4)$ to the x-axis:



$$\sqrt{1^2+4^2}$$
$$=\sqrt{17}.$$

Equation of a circle, center (a,b), radius r

$$(x-a)^2+(y-b)^2=r^2.$$

Equation of a sphere, center (a,b,c), radius r

$$(x-a)^2+(y-b)^2+(z-c)^2=r^2$$

Eg: Sphere with center $(1,2,3)$, radius 4.

$$\Rightarrow (x-1)^2+(y-2)^2+(z-3)^2=16.$$

Eg. Find the center and radius of the sphere $x^2 + y^2 + z^2 + 2x - 6z + 8 = 0$

$\Rightarrow (x+1)^2 + y^2 + (z-3)^2 = 2.$